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Empiric Antibiotic Guidelines for Skin and Soft Tissue Infections in Patients on Pediatric Services

This guideline is designed to provide guidance in pediatric patients with a primary skin and soft tissue infection (SSTI). Management of skin and soft tissue infections in patients < 2 months of age or those presenting with sepsis or septic shock not related to necrotizing fasciitis is beyond the scope of these guidelines. For sepsis or septic shock, refer to the Pediatric Sepsis Guidelines.

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Traumatic Wound Infections *WITH* Water Exposure	Footnotes	References

Setting	Empiric Therapy	Duration/Comments
Minor Skin Infections • Localized impetigo (non-bullous or bullous) • Secondarily infected skin lesions such eczema, ulcers, or lacerations • Folliculitis (small follicular abscess in epidermis) *Topical therapy:* Generally preferred over oral therapy *Oral therapy:* Indicated instead of topical therapy for patients with numerous impetigo lesions or in outbreak settings to reduce transmission *Target Pathogens:* *Staphylococcus aureus,* group A *Streptococcus*	Topical Therapy **Mupirocin 2%** topical ointment applied BID Oral Therapy *1st line:* **Cephalexin*** 25 mg/kg/DOSE PO TID (max: 1 g/DOSE) *If MRSA risk factors present[1] or allergy that precludes cephalexin use (see footnote[4]):* **TMP-SMX[2]*** 6 mg of TMP/kg/DOSE PO BID (max: 320 mg TMP/DOSE) Alternative to TMP-SMX[2] if sulfa allergy: **Clindamycin** 10 mg/kg/DOSE PO TID (max: 450 mg/DOSE)	Duration: 5 days *S. aureus* isolates from impetigo are commonly methicillin susceptible (MSSA). Michigan Medicine *S. aureus* resistance rates are low for TMP-SMX[2] (2%), compared to clindamycin (19% for MSSA and 25% for methicillin-resistant *S. aureus* [MRSA] in 2022). If worsening or not improving after 48 hours of oral cephalexin therapy, consider changing to an agent with anti-MRSA activity (i.e., TMP-SMX[2]).
Non-Purulent Cellulitis Absence of purulent drainage or exudate, ulceration, and no associated abscess. Includes erysipelas. *Target Pathogens:* Group A *Streptococcus, Staphylococcus aureus* (the role of community-acquired MRSA is unknown)	Outpatient or Step-down (from IV to PO) Therapy: *1st line:* **Cephalexin*** 25 mg/kg/DOSE PO TID (max: 1 g/DOSE) *If MRSA risk factors present[1] or allergy that precludes cephalexin use (4):* **TMP-SMX[2]*** 6 mg of TMP/kg/DOSE PO BID (max: 320 mg TMP/DOSE) Alternative to TMP-SMX[2] if sulfa allergy: **Clindamycin** 10 mg/kg/DOSE PO TID (max: 450 mg/DOSE) OR **Linezolid[8]** PO <12 years: 10 mg/kg/DOSE TID (max: 600 mg/DOSE) ≥12 years: 10 mg/kg/DOSE BID (max: 600 mg/DOSE) Inpatient (IV) Therapy *1st Line:* **Cefazolin*** 33 mg/kg/DOSE IV q8h (max: 2 g/DOSE) *Alternative if MRSA risk factors present[1] or allergy that precludes cefazolin use (4)* **Vancomycin** IV*	Duration: 5 days • May extend therapy up to 7-10 days if lack of symptom resolution at 5 days. Cephalexin and cefazolin provide coverage for group A *Streptococcus* and MSSA. TMP-SMX provides adequate coverage for group A *Streptococcus*, MSSA, and MRSA. If worsening or not improving after 48 hours of oral cephalexin therapy, consider changing to an agent with anti-MRSA activity (i.e., TMP-SMX[2] or linezolid). Linezolid suspension may not be readily available at all community pharmacies. Some insurance companies (including state Medicaid) may require prior authorization.

Setting	Empiric Therapy	Duration/Comments
Purulent Cellulitis or Abscesses including Folliculitis, Furuncles, Carbuncles *Abscess:* Collection of pus within the dermis and deeper skin tissues *Furuncle:* Infection of the hair follicle with suppuration extending through the dermis into subcutaneous tissue *Carbuncle:* Confluence of furuncles with wider infiltration *Target Pathogen:* *Staphylococcus aureus* (including MRSA)	*Incision and drainage (I&D) is recommended as primary management for abscesses. Antibiotics** are (at a minimum) recommended if patient meets one of the following criteria:* • Substantial surrounding cellulitis • Abscess >2 cm in diameter; >1 cm in infants and young children • Inability to adequately drain the abscess • Signs or symptoms of systemic illness (e.g., fever ≥38°C) • Immunodeficiency • Multiple sites Outpatient Therapy or Step-down (from IV to PO) Therapy *1st Line:* **TMP-SMX**[2,*] 6 mg of TMP/kg/DOSE PO BID (max: 320 mg TMP/DOSE) *Alternative for sulfa allergy:* **Doxycycline**[3] 2.2 mg/kg/DOSE PO BID (max: 100 mg/DOSE) Inpatient (IV) Therapy *1st Line:* **Vancomycin** IV* *Alternative for vancomycin allergy (not vancomycin infusion reaction):* **Linezolid**[8] PO/IV (PO preferred): < 12 years: 10 mg/kg/DOSE TID (max: 600 mg/DOSE) ≥ 12 years: 10 mg/kg/DOSE BID (max: 600 mg/DOSE)	Duration: 5 days • May extend therapy up to 7-10 days if lack of symptom resolution at 5 days. Cultures and susceptibilities are recommended when I&D is performed. Blood cultures are also recommended for patients with fever, rapidly progressive cellulitis, and systemic illness. Michigan Medicine *S. aureus* resistance rates are low for TMP-SMX[2] (2%) and doxycycline (3%), compared to clindamycin (19% for methicillin-susceptible *S. aureus* [MSSA] and 25% for methicillin-resistant *S. aureus* [MRSA] in 2022). *Tailor antibiotic therapy* to results of Gram stain, culture, and sensitivities. **Although ~70% of abscesses may resolve with I&D alone, an additional 10% are more likely to resolve with the addition of antibiotics. Clinical context should be taken into account when deciding if antibiotics are appropriate. Linezolid suspension may not be readily available at all community pharmacies. Some insurance companies (state Medicaid) may require prior authorization.

Setting	Empiric Therapy	Duration/Comments
Staphylococcal Scalded Skin Syndrome (SSSS) Results in loss of keratinocyte cell adhesion and leads to blistering of upper layer of the skin Pediatric Infectious Diseases consultation is recommended. Consider Dermatology consult if diagnosis is unclear or specific skin care recommendations are needed *Common pathogens:* *Staphylococcus aureus* (MSSA predominantly reported in the literature)	1st Line: **Cefazolin*** 33 mg/kg/DOSE IV q8h (max: 2 g/DOSE) + **Linezolid**[8] PO/IV (PO preferred): < 12 years: 10 mg/kg/DOSE TID (max: 600 mg/DOSE) ≥ 12 years: 10 mg/kg/DOSE BID (max: 600 mg/DOSE Step-down (from IV to PO) Therapy *1st Line:* **Cephalexin*** 25 mg/kg/DOSE PO TID (max: 1 g/DOSE) *Alternative if MRSA risk factors present or allergy that precludes cephalexin use (4):* **TMP-SMX**[2,*] 6 mg of TMP/kg/DOSE PO BID (max: 320 mg TMP/DOSE)	Duration: 10 days Consider discontinuing linezolid when patient is clinically stable (e.g., vital signs within normal limits, no vasopressor requirements) for 24-48 hours and rash no longer progressing (usual duration of 3-5 days). Staphylococcal Scalded Skin Syndrome (SSSS) is usually diagnosed in children <5 years of age.
Necrotizing Fasciitis Early and aggressive surgical exploration and debridement is critical. Emergent surgical consultation and ID consult are strongly recommended. *Common pathogens:* Group A β-hemolytic *Streptococcus*, *S. aureus,* *E. coli,* *Pseudomonas* spp., *Enterobacter* spp., *Klebsiella* spp., *Proteus* spp., *Bacteroides* spp., *Clostridia* spp., *Peptostreptococcus* spp.	1st Line: **Piperacillin-tazobactam*** 75 mg of piperacillin/kg/DOSE IV q6h (max: 4 g piperacillin/DOSE) extended infusion + **Vancomycin** IV* + **Clindamycin** 13 mg/kg/DOSE IV q8h (max: 900 mg/DOSE) Alternative for low-risk allergy[5] to penicillins: **Cefepime*** 50 mg/kg/DOSE IV q8h (max: 2 g/DOSE) extended infusion + **Vancomycin** IV* + **Clindamycin** 13 mg/kg/DOSE IV q8h (max: 900 mg/DOSE) ADD **Metronidazole** 10 mg/kg/DOSE PO/IV (PO preferred) TID (max: 500 mg/DOSE) if perineum or groin involved *Alternative for allergy that precludes use of both piperacillin-tazobactam and cefepime (4):* REPLACE cefepime with **Aztreonam*** 50 mg/kg/DOSE IV q8h (max: 2 g/DOSE) Alternative for vancomycin allergy (not vancomycin infusion reaction): **Piperacillin-tazobactam*** 75 mg of piperacillin/kg/DOSE IV q6h (max: 4 g piperacillin/DOSE) extended infusion + **Linezolid**[6] PO/IV (PO preferred): < 12 years: 10 mg/kg/DOSE TID (max: 600 mg/DOSE) ≥ 12 years: 10 mg/kg/DOSE BID (max: 600 mg/DOSE)	Duration: Empiric antibiotics should be continued until the following criteria are met: • Debridement no longer needed, • Clinical improvement, and • Minimum of 48-72 hours after completion of surgical debridement Clindamycin is initiated for anti-toxin activity for *Streptococcal* and *Staphylococcal* infections and can be stopped after 24-72 hours if infection has improved and patient is stable. *Tailor antibiotic therapy* to results of deep tissue Gram stain, culture, and sensitivities. Linezolid has in-vitro data that demonstrates suppression of toxin production with *S. aureus* and group A streptococcus. Clinical success against toxic shock syndrome is reported in case reports.

Setting	Empiric Therapy	Duration/Comments
Traumatic Wound Infections *WITHOUT* Water Exposure Usually polymicrobial from environmental contamination. See section above if concern for necrotizing fasciitis. For animal/human bites, refer to Animal Bite Guidelines on antimicrobial stewardship webpage. Evaluate tetanus immunization status, and if indicated, administer tetanus immunization +/- tetanus immune globulin. *Target pathogens:* *Staphylococcus aureus*, *Clostridia* spp., *Bacteroides* spp., *Prevotella* spp., *Porphyromonas* spp., *Peptostreptococcus* spp.	Traumatic wounds without evidence of local infection or systemic signs of infection typically do not need antimicrobial therapy. Outpatient (PO) Therapy *1st Line:* **Amoxicillin-clavulanate*** 25 mg amoxicillin/kg/DOSE PO BID (max: 875 mg amoxicillin/DOSE) 7:1 formulation is recommended (400/57/ 5ml or 200/28.5/5 ml) If MRSA risk factors present[1] ADD **TMP-SMX**[2,*] 6 mg of TMP/kg/DOSE PO BID (max: 320 mg TMP/DOSE) *Alternative for low-risk allergy[5] to penicillins:* **Cephalexin*** 25 mg/kg/DOSE PO TID (max: 1 g/DOSE) + **Metronidazole** 10 mg/kg/DOSE PO TID (max: 500 mg/DOSE) *Alternative for allergy that precludes use of both amoxicillin-clavulanate and cephalexin (4):* **TMP-SMX**[2,*] 6 mg of TMP/kg/DOSE PO BID (max: 320 mg TMP/DOSE) + **Metronidazole** 10 mg/kg/DOSE PO TID (max: 500 mg/DOSE) Inpatient (IV) Therapy *1st Line:* **Ampicillin-sulbactam*** 50 mg of ampicillin/kg/DOSE IV q6h (max: 2 g ampicillin/DOSE) *Alternative for low-risk allergy[5] to penicillins:* **Cefazolin*** 33 mg/kg/DOSE IV q8h (max: 2 g/DOSE) + **Metronidazole** 10 mg/kg/DOSE PO/IV (PO preferred) TID (max: 500 mg/DOSE) *Alternative if MRSA risk factors present[1], or allergy that precludes use of both ampicillin-sulbactam and cefazolin (4):* **Vancomycin IV*** + **Metronidazole** 10 mg/kg/DOSE PO/IV (PO preferred) q8h (max: 500 mg/DOSE)	Duration: 7 days • Therapy may need to be extended based on severity of infection and response to treatment. Consider Pediatric ID consult for infections that are deep, extensive or respond slowly Debridement of devitalized tissues and contaminating debris is critical to source control and successful healing. Empiric therapy should take into account site of wound and prior cultures and colonization. *Tailor antibiotic therapy* to results of deep tissue Gram stain, culture, and sensitivities.

Setting	Empiric Therapy	Duration/Comments
Traumatic Wound Infections *WITH* Water Exposure Usually polymicrobial from environmental contamination. See section above if concern for necrotizing fasciitis. For animal/human bites, refer to Animal Bite Guidelines on antimicrobial stewardship webpage. Evaluate tetanus immunization status, and if indicated, administer tetanus immunization ± tetanus immune globulin. *Target pathogens:* *Staphylococcus aureus*, *Clostridia* spp., *Bacteroides* spp., *Prevotella* spp., *Porphyromonas* spp., *Peptostreptococcus* spp. Consider *Aeromonas* and *Pseudomonas* spp., other gram negatives if significant water exposure	Outpatient (PO) Therapy: **Levofloxacin*** PO: < 5 years: 10 mg/kg/DOSE PO BID (max: 375 mg/DOSE) ≥ 5 years: 10 mg/kg/DOSE PO daily (max: 750 mg/DOSE) + **Metronidazole** 10 mg/kg/DOSE PO TID (max: 500 mg/dose) If MRSA risk factors present[1] ADD **TMP-SMX**[2,*] 6 mg of TMP/kg/DOSE PO BID (max: 320 mg TMP/DOSE) Inpatient (IV) Therapy: *1st Line:* **Cefepime*** 50 mg/kg/DOSE IV q8h (max: 2 g/DOSE) extended infusion + **Metronidazole** 10 mg/kg/DOSE PO/IV (PO preferred) q8h (max: 500 mg/DOSE) If MRSA risk factors present[1] ADD **Vancomycin** IV* *Alternative for allergy that precludes cefepime use[4]:* **Levofloxacin** IV/PO (PO preferred): < 5 years: 10 mg/kg/DOSE PO BID (max: 375 mg/DOSE) ≥ 5 years: 10 mg/kg/DOSE PO daily (max: 750 mg/DOSE) + **Metronidazole** 10 mg/kg/DOSE PO/IV TID (PO preferred) (max: 500 mg/DOSE) If MRSA risk factors present[1] ADD **Vancomycin** IV*	Duration: 7 days • Therapy may need to be extended based on severity of infection and response to treatment. Consider Pediatric ID consult for infections that are deep, extensive or respond slowly Debridement of devitalized tissues and contaminating debris is critical to source control and successful healing. Empiric therapy should take into account site of wound and prior cultures and colonization. *Vibrio vulnificus* wound infections require extensive debridement and mortality can be high. Consider combination therapy with ceftazidime and doxycycline. *Tailor antibiotic therapy* to results of deep tissue Gram stain, culture, and sensitivities.

Footnotes:

* Renal adjustment may be necessary. See Pediatric Antimicrobial Dosing Guidelines.

[1] Consider MRSA coverage if any of the following are present: severe sepsis or septic shock, immunocompromised status, personal or household contact with MRSA infection, or colonization in the past 12 months

[2] TMP-SMX = trimethoprim-sulfamethoxazole

[3] CDC and Indian Health Service (IHS) study demonstrated short courses (7-10 days) of doxycycline can be used in children without causing tooth staining or weakening of tooth enamel. Todd SR et al. J Pediatr. 2015;166(5):1246-1251.

[4] See β-lactam allergy evaluation and empiric guidance for further information.

[5] Low-risk allergies include: pruritus without rash, remote (>10 years) unknown reaction, patient denies allergy but is on record, mild rash with no other symptoms (mild rash: non-urticarial rash that resolves without medical intervention). See β-lactam allergy evaluation and empiric guidance for further information.

[6] Serotonin Syndrome and Linezolid: Education and Recommendations

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Revision History

Date	**Revision**
04/2020	Reduced some clindamycin doses to align with adult SSTI and animal bite guideline dosing; updated allergy wording
09/2020	Adjusted aztreonam dosing
03/2021	Updated vancomycin hyperlinks
09/2021	Updated vancomycin infusion reaction terminology

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source.